PROJECT INITIATION DOCUMENT

CYBERNOVA

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Table of Contents

Section 1: Problem definition 2
1.Description of the organization/ environment 2
2.Current process used by the organisation 2
3.Operational problems experienced 2
4.The consequences of the current problem 2
5.Why a digital Information system could improve the situation 3
SECTION 2: STAKEHOLDER IDENTIFICATION AND PERSONAS 3
1.Identify The Stakeholder 3
2.Describe their role in the organisation 3
3. Explain how they interact with the system. 4
4. Personas 5
SECTION 3: SYSTEM OBJECTIVES 6
SECTION 4: WORK BREAKDOWN STRUCTURE (WBS) 6
1. Project Planning 6
2. Requirements Analysis 7
3. System Design 7
4. System Development 8
5. Testing 8
6. Final Demonstration Preparation 8
Conclusion 9

Section 1: Problem definition

1.Description of the organization/ environment

The company provides bus transportation services in Forte Town. Its primary functions are managing a fleet of buses, planning and carrying out routes, scheduling drivers, managing ticketing and passenger tracking, and supervising the vital aspects of fuel usage and vehicle maintenance. This service is essential for linking different areas of Forte Town and guaranteeing that residents and commuters have dependable access to transportation.

2.Current process used by the organisation

The company now uses a lot of manual systems, such as:

- Logbooks on paper for tracking routes and travel
- Using manual fuel logs to monitor fuel usage
- Buses with handwritten maintenance records
- Scheduling drivers manually
- Paper-based or unofficial passenger tracking and ticketing

3.Operational problems experienced

Several operational issues have arisen from the use of manual systems:

- Route errors brought on by inadequate coordination and ambiguous records
- Bus failures resulting from poor or postponed maintenance
- Inaccurate fuel monitoring that could result in overuse or inefficiency
- Conflicts over driver scheduling and ineffective time management
- Ticket sales and passenger tracking are challenging
- Handwritten records can cause data mistakes and information loss
- Slow communication between managements, drivers and employees

4.The consequences of the current problem

The organization is badly impacted by these issues in several ways:

- Unreliable transportation services that cause passenger delays
- Higher running costs because of fuel waste and unforeseen repairs

- Decreased client trust and pleasure
- Staff productivity is reduced because of ineffective workflows
- Poor decision-making due to insufficient or incorrect data
- Revenue loss due to ineffective ticket monitoring and inefficient systems

5.Why a digital Information system could improve the situation

Putting in place a digital information system would greatly enhance operations by:

- Automating route management and bus fleet registration
- Reducing conflicts and enhancing driver scheduling
- Facilitating precise ticketing and passenger monitoring
- Monitoring fuel and tracking maintenance in real time
- Minimizing data mistakes and avoiding information loss
- Enhancing cooperation and communication within the company
- Producing helpful reports like:
    1. Summaries of daily route completion
    2. Trends in fuel efficiency
    3. Reports on driver performance
    4. Alerts for maintenance (due/overdue)

SECTION 2: STAKEHOLDER IDENTIFICATION AND PERSONAS

1.Identify The Stakeholder

- Fleet Manager
- Bus Drivers
- Admin
- Maintenance staff
- Passengers
- Ticket Conductor

2.Describe their role in the organisation

- Fleet Manager- They monitor buses, plan and manage route for bus drivers.
- Bus Drivers- Follow instructions from the fleet managers about routes and make sure they report fuel and maintenance issues. Transport passengers safely

- Admin- Manages the system, ensures it runs smoothly, and controls user access.
- Maintenance Staff-Ensure buses are in good working condition by servicing them every month.
- Passengers- They use buses for transportation
- Ticket Conductors-The conductor is responsible for issuing tickets, checking passengers, and recording passenger information using the system instead of manual methods.

3. Explain how they interact with the system.

Fleet Manager:

-Registers buses, creates and assigns routes, and monitor bus operations

- Reviews reports on route completion and fuel efficiency

Bus Driver:

-Access the system to view assigned routes and schedules

- Report fuel usage and maintenance issues using the system.

- Update trip status to indicate completion of routes.

Passenger:

-Access the system online to view bus routes, schedules, and availability.

-Purchase the ticket and receive confirmation email.

Ticket Conductor:

- Use system terminals to issue for passengers.

- Record passenger numbers per route

- Provide data for route demand analysis

Maintenance Staff:

- Uses the system to view the maintenance schedule for buses.

- Log repairs and servicing activities

-Update vehicle condition status

- Receives alerts for upcoming or overdue services to ensure buses remain operational.

Admin:

- Set up system settings

-Manage user accounts (drivers, managers, staff)

-Ensure data integrity and backups

-Monitor system performance

4. Personas

Name	Age	Role	Technology Experience Level	Goals when using the system	Problems they currently experience
1.Buhle Zulu	30	Fleet Manager	Intermediate	Monitor bus performance.	Current manual logs are time-consuming and likely to contain mistakes.
2.Luzuku Pantsi	33	Ticket Conductor	Advanced	Accurately record the number of passengers.	Ticket errors, manual recording leads to mistakes.
3.Ashley Witbooi	27	Maintenance Staff	Beginner	Ensure buses are in good working condition.	Hard to track which buses have upcoming or overdue maintenance

SECTION 3: SYSTEM OBJECTIVES

The proposed system aims to improve the overall efficiency, accuracy, and reliability of the bus transportation operations in Forte Town. The objectives of the system are as follows:

1. **To improve the accuracy of route management**
   This will reduce route errors and ensure that buses operate according to correct and well-coordinated schedules.
2. **To reduce conflicts in driver scheduling**
   This will ensure that drivers are assigned efficiently and that time management is improved.
3. **To improve the monitoring of fuel usage**
   This will help reduce fuel waste and ensure better control over fuel consumption.
4. **To ensure timely and effective vehicle maintenance**
   This will reduce the likelihood of unexpected bus breakdowns and improve service reliability.
5. **To improve the accuracy of ticketing and passenger tracking**
   This will ensure that all passenger data and ticket sales are recorded correctly.
6. **To reduce data errors and prevent loss of information**
   This will improve the reliability of organisational data and record-keeping.
7. **To improve communication between management, drivers, and staff**
   This will support faster decision-making and better coordination within the organisation.
8. **To improve decision-making through access to accurate information**
   This will enable management to make informed decisions based on reliable operational data.

SECTION 4: WORK BREAKDOWN STRUCTURE (WBS)

1. Project Planning

1.1 Define project scope and objectives

1.2 Identify stakeholders (bus company, drivers, passengers, maintenance staff)

1.3 Develop project schedule and milestones

1.4 Allocate resources and responsibilities

1.5 Risk assessment and mitigation planning

2. Requirements Analysis

2.1 Gather requirements from stakeholders

2.2 Document functional requirements

→ Bus fleet registration

→ Route creation & assignments

→ Driver scheduling

→ Ticketing & passenger tracking

→ Fuel & maintenance logging

2.3 Define reporting needs

→ Daily route completion summary

→ Fuel efficiency trends

→ Driver performance report

→ Maintenance due/overdue list

2.4 Validate requirements with stakeholders

3. System Design

3.1 Database design

→ Fleet database schema

→ Route & scheduling tables

→ Ticketing & passenger tracking tables

→ Fuel & maintenance logs

3.2 Interface design

→ Admin dashboard

→ Driver mobile interface

→ Passenger ticketing interface

3.3 Workflow design

→ Route assignment workflow

→ Driver scheduling workflow

→ Maintenance logging workflow

3.4 Security & access control design

4. System Development

4.1 Backend development

→ Database implementation

→ API development for fleet, routes, drivers, tickets, fuel logs

4.2 Frontend development

→ Web portal for administrators

→ Mobile app for drivers

→ Passenger ticketing interface

4.3 Integration of modules

→ Fleet registration with route assignments

→ Driver scheduling with ticketing system

→ Fuel & maintenance logging with reporting engine

5. Testing

5.1 Unit testing (individual modules)

5.2 Integration testing (cross-module workflows)

5.3 User acceptance testing (drivers, admins, passengers)

5.4 Performance testing (route completion, fuel efficiency tracking)

5.5 Security testing (data protection, access control)

6. Final Demonstration Preparation

6.1 Prepare demonstration scripts (fleet registration, route assignment, ticketing, maintenance logging)

6.2 Generate sample reports (daily route summary, fuel trends, driver performance, maintenance list)

6.3 Conduct stakeholder walkthrough

6.4 Collect feedback and finalize system documentation

Conclusion

The current manual processes used by the bus transportation company in Forte Town have resulted in several operational inefficiencies, including inaccurate data recording, poor communication, scheduling conflicts, and unreliable service delivery. These challenges negatively impact both the organisation and its passengers by increasing costs, reducing productivity, and lowering customer satisfaction.

The proposed Bus Transportation Management System aims to address these issues by introducing a digital solution that automates key operations such as route management, driver scheduling, ticketing, fuel monitoring, and maintenance tracking. By improving data accuracy, enhancing communication, and enabling real-time reporting, the system will support better decision-making and more efficient workflows.

Overall, the implementation of this system is expected to significantly improve service reliability, reduce operational costs, and enhance the overall performance of the organisation. This project provides a strong foundation for developing a practical and effective information system that meets the needs of all stakeholders involved.